



AASHINI RAJPAL

RESEARCH ENGINEER

Chemist/physicist with a strong foundation in experimental design, spectroscopy, microfluidics, dosimetry and process optimization



[Portfolio](#)



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SKILLS

- Simulations (*GAUSSIAN, CPMD, BOMD, COMSOL, Chemsimul*)
- Spectroscopy techniques
- Microfluidics
- Proposal Writing
- Analysis softwares (*IGOR pro, CasaXPS, SpecsLab Prodigy, OxyGEN, ChemDraw, OceanView, Avantes*)
- Python (Intermediate)

EDUCATION

2020 - 2023

PhD in Chemistry

University Paris-Saclay

2018 - 2020

Master in Physical Chemistry

Erasmus Mundus SERP+ Joint Master Degree, France and Poland (16.5/20)

LANGUAGES

English ★ ★ ★ ★

Hindi ★ ★ ★ ★

French ★ ★ ★

REFERENCE

CH.ROBERT@gustaveroussy.fr

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WORK EXPERIENCE

- **Gustave Roussy Cancer Institute and Theryq** 2024 - PRESENT
Research Engineer
 - Performed dosimetric characterization under ultra-high dose-rate (UHDR) conditions using multiple techniques.
 - Designed and optimized experimental workflows to investigate the FLASH/UHDR effect.
 - Investigated chemistry-driven mechanisms underlying differential UHDR biological responses.
 - Conducted heat diffusion simulations using COMSOL.
 - Collaborated on projects integrating chemistry, physics, and radiobiology.
- **SOLEIL synchrotron, Sorbonne University and CEA Saclay** 2020 - 2023
Research scholar
 - Developed and optimized a highly sensitive in-line analysis setup integrated with microfluidic devices.
 - Investigated light-matter interaction using experiments and simulations. Worked with electron and photon radiation sources.
 - Led a team of 5 researchers for 10 intensive experimental projects.
 - Presented research findings at national and international conferences.
- **Indian Institute of Technology Bombay (IIT B)** 2017
Research scholar
 - Computationally studied reaction mechanisms to analyze activation energies and molecular structures

CERTIFICATIONS

- **Good Clinical Practices**, National Institute on Drug Abuse (NIDA) Center for Clinical Trials (CCTN) Clinical Trials Network (CTN)
- **Project designer - Regulatory course on animal experiment**, approved by the Ministry of Agriculture
- **Molecule to Market job simulation**, Pfizer UK.
- **Basics of AI in Medical Physics**, Gustave Roussy

PUBLICATIONS

[Researchgate/aashinirajpal](https://www.researchgate.net/profile/Aashini-Rajpal)